Project 1 — Microsoft Model Validation and AI-Use Record
Company: Microsoft Corporation (MSFT)
Valuation date: September 1, 2026
Prepared: October 19, 2026
Investment recommendation: WATCH / DEFER

1. Purpose and scope

This document records the model outputs, source checks, sensitivity tests, validation observations, and AI-use disclosure supporting the Edition B research and investment committee materials. It distinguishes observed checks from outstanding verification. It is not a claim that all inputs or methods have been independently audited.

2. Reproducible model outputs

Model	Script	Base-case result
FCFF discounted cash flow	dcf.py	$197.71/share
FCFE pro-forma valuation	msft_proforma.py	$116.94/share
Two-peer median P/E	comps.py	$371.30/share
FCFF / FCFE comparison difference	project1_reconciliation.py	$80.77/share
Interactive workbench	app.py	Displays both intrinsic estimates and scenario controls

The valuation-date Microsoft reference price is $501.02/share. The three valuation methods are different scenarios and should not be averaged into a single target. Their base-case estimates are below the reference price.

3. Historical data and source checks

The Microsoft FY2026 Form 10-K was used to check operating cash flow of $182,935 million, capital expenditures of $115,948 million, interest expense of $3,051 million, and tax inputs used in the after-tax interest calculation. The revised starting FCFF is:

$182,935m + $2,459.2199m − $115,948m = $69,446.2199m.

The earlier starting FCFF was $69,408.2199m. Correcting the operating cash flow and capital expenditure inputs increased the FCFF valuation from approximately $197.60/share to $197.71/share. The Microsoft closing price of $501.02 on September 1, 2026 was checked against Yahoo Finance historical data.

Outstanding check: The Oracle and Adobe historical share prices and EPS measurement conventions in the P/E model have not been independently verified for consistent valuation-date treatment. Their peer-implied result remains a provisional cross-check.

4. Accounting and numerical validation

The base-case pro-forma model passed its reported balance-sheet consistency checks. The FCFF, FCFE, peer-comparison, and reconciliation outputs were reproduced using the project Python scripts. These checks support internal reproducibility; they do not establish that the FCFF and FCFE models are economically equivalent.

The FCFF DCF uses a 10% WACC, 3% terminal growth rate, and five-year FCFF growth path of 15%, 13%, 11%, 9%, and 7%. Approximately 74.28% of modeled enterprise value comes from terminal value, making the

estimate sensitive to long-run assumptions.

5. Sensitivity and failure/recovery tests

Scenario	Observed or reported result
FCFF: WACC 11%, terminal growth 2%	$158.60/share
FCFF: WACC 10%, terminal growth 3%	$197.71/share
FCFF: WACC 9%, terminal growth 4%	$267.78/share
FCFE: higher revenue-growth path (+2 percentage points each forecast year)	Approximately $144.44/share
FCFE: gross margin raised from 68% to 70%	Approximately $131.18/share
Streamlit minimum cash requirement raised to $100,000m	Validation warning displayed
Minimum cash requirement returned to $0	Base-case configuration restored

All nine tested WACC/terminal-growth combinations in the Edition B grid are below the $501.02 reference price. The reverse-DCF search did not find a solution within its configured uniform growth-shift range of −5 to +10 percentage points; this is a limitation of that search range, not proof that the market price is impossible to justify.

6. Model limitations and unresolved work

- Different cash-flow approaches: The FCFF model forecasts FCFF directly and bridges enterprise value to equity; the FCFE model forecasts equity cash flows from simplified financial projections. Operating forecasts, financing flows, discount rates, and nonoperating adjustments have not been fully reconciled.
- Terminal-value exposure: The DCF is heavily dependent on the terminal assumption.
- Comparable-company limitations: Only Oracle and Adobe are used; peer selection materially changes the implied price, and price/EPS inputs require further independent verification.
- Share count and bridge simplifications: FY2026 diluted weighted-average shares are used as a working denominator, not actual shares outstanding on the valuation date. The enterprise-to-equity bridge does not separately adjust for all possible nonoperating claims.
- Forecast uncertainty: AI capital expenditures, cloud growth, margins, and returns on infrastructure spending are uncertain.

7. AI-use disclosure and analyst responsibility

AI assistance supported research organization, Python troubleshooting, code review, model documentation, drafting, and presentation preparation. AI-generated explanations and calculations were compared with the available model outputs where possible. The analyst is responsible for primary-source verification, model assumptions, financial interpretation, investment judgment, and the final recommendation. Any outstanding verification is disclosed above rather than represented as completed.

8. Reproduction instructions

From the TICKER-research repository with the project's Python environment activated, run:

```
python dcf.py
python msft_proforma.py
python comps.py
python project1_reconciliation.py
python -m streamlit run app.py
```

In the Streamlit app, inspect the base-case values, change gross margin from 68% to 70%, restore 68%, then set the minimum cash requirement to $100,000m and restore it to $0. Record the observed warnings and outcomes in the validation video and corrected transcript. The actual video and transcript are separate submission artifacts and are not included in this document.

9. Supporting sources and files

- Microsoft Corporation, FY2026 Form 10-K: https://www.sec.gov/Archives/edgar/data/789019/000119312526323660/msft-20260630.htm
- Yahoo Finance, MSFT historical prices: https://finance.yahoo.com/quote/MSFT/history/
- Project_1_Edition_B_MSFT.md
- project1_source_verification.md
- project1_valuation_reconciliation.md
- project1_validation.md
- 2026-09-10_MSFT_DCF_Inputs.md
- dcf.py, msft_proforma.py, comps.py, project1_reconciliation.py, app.py